Average-case generalized-inner-product lower bounds for formulas with low-communication leaf gates

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Editorial note. Editable transcription only of corresponding formalCCBY3.0 Unported content; nonexclusive nativev1 is private aid. EntireSection3 andAppendixA.1 proofs retained with requiredSection2 definitions/cited prior facts. Original author probabilities/inequalities/constants retained without mathematical repair or new assumptions. MainTheorem1Items2/3 are externaloriginal scope references, not imported or counted here. Author metadata unchanged; title-page author list split into two lines to fit. Native unsupported/unused publisher packages omitted, necessarymath macros explicit, no input/include/assets. All production/validationYupeng; root owns compilation.

Allowed standard prerequisites (editorial).

- DeMorgan formula approximation, multiparty randomized/deterministic communication and GIP communication lower bound. FormalSection2 model/approximation statements and citedBNS lowerbound; source-local correlation conversion and fullTal-formula lemma proofs inAppendixA.1 included

GIP_n^k**.** The Generalized Inner Product function $\mathsf{GIP}_n^k\colon \{0,1\}^n \to \{0,1\}$ is defined as

$$\mathsf{GIP}_n^k\left(x^{(1)}, x^{(2)}, \ldots, x^{(k)}\right) = \sum_{j=1}^{n/k} \bigwedge_{i=1}^{k} x_j^{(i)} \pmod 2,$$

where $x^{(i)} \in \{0,1\}^{n/k}$ for each $i \in [k]$.

2 Preliminaries

2.1 Notation

Let $n \in \mathbb{N}$; we denote $\{1, \ldots, n\}$ by $[n]$, and denote by U_n the uniform distribution over $\{0,1\}^n$. We use $\widetilde{O}(\cdot)$ (and $\widetilde{\Omega}(\cdot)$) to hide polylogarithmic factors. That is, for any $f\colon \mathbb{N} \to \mathbb{N}$, we have that $\widetilde{O}(f(n)) = O(f(n) \cdot \mathrm{polylog}(f(n)))$.

In this paper, we will mainly use $\{-1,1\}$ as the Boolean basis. In some parts of this paper, we will use the $\{0,1\}$ basis for the simplicity of the presentation. This will be specified in corresponding sections.

2.2 De Morgan formulas and extensions

Definition 8. *An n-variate* de Morgan formula *is a directed rooted tree; its non-leaf vertices (henceforth,* internal gates*) take labels from $\{\mathrm{AND}, \mathrm{OR}, \mathrm{NOT}\} = \{\wedge, \vee, \neg\}$ and its leaves (henceforth,* variable gates*) take labels from the set of variables $\{x_1, \ldots, x_n\}$. Each internal gate has bounded in-degree (henceforth,* fan-in*); the* NOT *gate in particular has fan-in* 1 *and every variable gate has fan-in* 0*. The* size *of a de Morgan formula is the number of its leaf gates.*

In this work, we denote by $\mathsf{FORMULA}[s]$ the class of Boolean functions computable by size-s de Morgan formulas. Let $\mathcal{G}$ denote some class of Boolean functions; then, we denote by $\mathsf{FORMULA}[s] \circ \mathcal{G}$ the class of functions computable by some size-s de Morgan formula where its leaves are labelled by functions in $\mathcal{G}$.

2.3 Approximating polynomials

Definition 9 (Point-wise approximation)**.** *For a Boolean function $f\colon \{-1,1\}^n \to \{-1,1\}$, we say that the function $\tilde{f}\colon \{-1,1\}^n \to \mathbb{R}$ ε-approximates f if for every $z \in \{-1,1\}^n$,*

$$\left| f(z) - \tilde{f}(z) \right| \leq \varepsilon.$$

We will need the following powerful result for the approximating degree of de Morgan formulas.

Theorem 10 ([4], see also [1])**.** *Let $s > 0$ be an integer and $0 < \varepsilon < 1$. Any de Morgan formula $F\colon \{-1,1\}^n \to \{-1,1\}$ of size s has a ε-approximating polynomial of degree $d = O(\sqrt{s} \cdot \log(1/\varepsilon))$. That is, there exists a degree-d polynomial $p\colon \{-1,1\}^n \to \mathbb{R}$ over the reals such that for every $z \in \{-1,1\}^n$,*

$$|p(z) - F(z)| \leq \varepsilon.$$

Note that Theorem 10 still holds if we use $\{0,1\}$ as the Boolean basis.

2.4 Communication complexity

We use standard definitions from communication complexity. In this paper we consider the standard two party model of Yao and its generalizations to multiparty setting. We denote deterministic communication complexity of a Boolean function by $D(f)$ in the two party setting. We refer to [3] for standard definitions from communication complexity.

Definition 11. *Let $f\colon \{0,1\}^n \to \{0,1\}$ be a Boolean function. The* communication matrix of f*, namely M_f, is a $2^{n/2} \times 2^{n/2}$ matrix defined by $(M_f)_{x,y} := f(x,y)$.*

Definition 12. *A* rectangle *is a set of the form $A \times B$, for $A, B \subseteq \{0,1\}^n$. A* monochromatic rectangle *is a rectangle S such that for all pairs $(x,y) \in S$ the value $f(x,y)$ is the same.*

Lemma 13. *Let* Π *be a protocol that computes* $f\colon \{0,1\}^n \to \{0,1\}$ *with at most* D *bits of communication. Then,* Π *induces a partition of* M_f *into at most* 2^D *monochromatic rectangles.*

Given a protocol, its *transcript* is the sequence of bits communicated.

Lemma 14. *For every transcript* z *of some communication protocol, the set of inputs* (x, y) *that generate* z *is a rectangle.*

Below, we recount the definitions of two multiparty communication models used in this work, namely the number-on-forehead and the number-in-hand models.

Definition 15 ("Number-on-forehead" communication model; informal)**.** *In the* k*-party "number-on-forehead" communication model, there are* k *players and* k *strings* $x_1, \ldots, x_k \in \{0,1\}^{n/k}$ *and player* i *gets all the strings except for* x_i*. The players are interested in computing a value* $f(x_1, \ldots, x_k)$*, where* $f\colon \{0,1\}^n \to \{0,1\}$ *is some fixed function. We denote by* $D^{(k)}(f)$ *the number of bits that must be exchanged by the best possible number on forehead protocol solving* f*.*

We also use the following weaker communication model.

Definition 16 ("Number-in-hand" communication model; informal)**.** *In the* k*-party "number-in-hand" communication model, there are* k *players and* k *strings* $x_1, \ldots, x_k \in \{0,1\}^{n/k}$ *and player* i *gets only* x_i*. The players are interested in computing a value* $f(x_1, \ldots, x_k)$*, where* $f\colon \{0,1\}^n \to \{0,1\}$ *is some fixed function. We denote by* $D^{(k\text{-NIH})}(f)$ *the number of bits that must be exchanged by the best possible communication protocol.*

Note that $D^{(k\text{-NIH})}(f) \leq (1 - 1/k) \cdot n + 1$, for any n-variate Boolean function f, as if $k-1$ players write on the blackboard their string, then the player that did not reveal her input may compute $f(x_1, \ldots, x_k)$ on her own and then publish it.

For the communication models mentioned above, there are also bounded-error randomized versions, denoted by R_δ, $R_\delta^{(k)}$, and $R_\delta^{(k\text{-NIH})}$, respectively, where $0 < \delta < 1$ is an upper bound on the error probability of the protocol. In this setting, the players have access to some shared random string, say r, and the aforementioned error probability of the protocol is considered over the possible choices of r. Moreover, we require the error to be at most δ on each fixed choice of inputs.

We can extend the definitions of the communication complexity measures, defined above, to classes of Boolean functions, in a natural way. That is, for any communication complexity measure $M \in \left\{D, D^{(k)}, D^{(k\text{-NIH})}, R_\delta, R_\delta^{(k)}, R_\delta^{(k\text{-NIH})}\right\}$ and for any class of Boolean functions $\mathcal{G}$, we may define

$$M(\mathcal{G}) := \max_{g \in \mathcal{G}} M(g)\,.$$

We note that throughout this paper, we denote by n the number of input bits for the function regardless the communication models. In the k-party communication setting (either NOF or NIH), we assume without loss of generality that n is divisible by k.

3 Lower bounds

In this section, we prove an average-case lower bound for the generalized inner product function against $\mathsf{FORMULA} \circ \mathcal{G}$, where $\mathcal{G}$ is the set of functions that have low-cost randomized communication protocols in the number-on-forehead setting. This corresponds to Item 1 of Theorem 1 of the original publication. Items 2 and 3 rely on our PRG constructions, and the proofs are deferred to Section 4 of the original publication.

Theorem 19. *For any integer $k \geq 2$, $s > 0$ and any class of functions $\mathcal{G}$, let $C\colon \{-1,1\}^n \to \{-1,1\}$ be a function in $\mathsf{FORMULA}[s] \circ \mathcal{G}$ such that*

$$\mathop{\mathbf{Pr}}_{x \sim \{-1,1\}^n} \left[C(x) = \mathsf{GIP}_n^k(x) \right] \geq 1/2 + \varepsilon.$$

Then

$$s = \Omega\left(\frac{n^2}{k^2 \cdot 16^k \cdot \left(R^{(k)}_{\varepsilon/(2n^2)}(\mathcal{G}) + \log n \right)^2 \cdot \log^2(1/\varepsilon)} \right).$$

We need a couple useful lemmas from [5], whose proofs are presented in Section A.1 (Theorem 50 and Theorem 51) for completeness.

Lemma 20 ([5])**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$, and let $f, C\colon \{-1,1\}^n \to \{-1,1\}$ be such that*

$$\mathop{\mathbf{Pr}}_{x \sim \mathcal{D}}[C(x) = f(x)] \geq 1/2 + \varepsilon.$$

Let $\tilde{C}\colon \{-1,1\}^n \to \mathbb{R}$ be a ε-approximating function of C, i.e., for every $x \in \{-1,1\}^n$, $|C(x) - \tilde{C}(x)| \leq \varepsilon$. Then,

$$\mathop{\mathbf{E}}_{x \sim \mathcal{D}}[\tilde{C}(x) \cdot f(x)] \geq \varepsilon.$$

Lemma 21 ([5])**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$ and let $\mathcal{G}$ be a class of functions. For $f\colon \{-1,1\}^n \to \{-1,1\}$, suppose that $D\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{FORMULA}[s] \circ \mathcal{G}$ is such that*

$$\mathop{\mathbf{Pr}}_{x \sim \mathcal{D}}[D(x) = f(x)] \geq 1/2 + \varepsilon_0.$$

Then there exists some $h\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{XOR}_{O\left(\sqrt{s} \cdot \log(1/\varepsilon_0)\right)} \circ \mathcal{G}$ such that

$$\mathop{\mathbf{E}}_{x \sim \mathcal{D}}[h(x) \cdot f(x)] \geq \frac{1}{s^{O\left(\sqrt{s} \cdot \log(1/\varepsilon_0)\right)}}.$$

We also need the following communication-complexity lower bound for GIP.

Theorem 22 ([2, Theorem 2])**.** *For any $k \geq 2$, any function that computes GIP_n^k on more than $1/2 + \delta$ fraction of the inputs (over uniformly random inputs) must have k-party deterministic communication complexity at least $\Omega\big(n/(k \cdot 4^k) - \log(1/\delta)\big)$.*

We first show that if a function correlates with some small formula, whose leaves are functions with low *randomized* communication complexity, then it also correlates non-trivially with some function of relatively low *deterministic* communication complexity.

Lemma 23. *For any distribution $\mathcal{D}$ over $\{-1,1\}^n$, and any class of functions $\mathcal{G}$, let $f\colon \{-1,1\}^n \to \{-1,1\}$ and $C\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{FORMULA}[s] \circ \mathcal{G}$ be such that*

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x) = f(x)] \geq 1/2 + \varepsilon.$$

Then there exists a function h, with k-party deterministic communication complexity at most

$$O\left(R^{(k)}_{\varepsilon/(2s)}(\mathcal{G}) \cdot \sqrt{s} \cdot \log(1/\varepsilon)\right),$$

such that

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[h(x) = f(x)] \geq 1/2 + 1/s^{O(\sqrt{s}\cdot\log(1/\varepsilon))}.$$

Proof. Let $C = F(g_1, g_2 \ldots, g_s)$ be the function in $\mathsf{FORMULA}[s] \circ \mathcal{G}$, where F is a formula and $g_1, g_2, \ldots, g_s$ are leaf functions from the class $\mathcal{G}$. For each g_i, consider a k-party randomized protocol Π_i of cost at most $R = R^{(k)}_{\varepsilon/(2s)}(\mathcal{G})$ that has an error $\varepsilon/(2s)$. Now consider the following function

$$\tilde{C}(x) := \mathop{\mathbf{E}}_{\Pi_1,\Pi_2,\ldots,\Pi_s}[D(x)],$$

where

$$D(x) := F(\Pi_1(x), \Pi_2(x), \ldots, \Pi_s(x)).$$

Note that for any fixed choice of $(\Pi_1, \Pi_2, \ldots, \Pi_s)$, D is a formula whose leaves are functions with *deterministic* communication complexity at most R. Next, we show the following.

Claim 24. *The function $\tilde{C}$ ε-approximates C.*

Proof of Theorem 24. First note that since each Π_i is a $(\varepsilon/(2s))$-error randomized protocol, by taking the union bound over the s leaf functions, we have that for every input $x \in \{-1,1\}^n$,

$$\mathop{\mathbf{Pr}}_{\Pi_1,\Pi_2,\ldots,\Pi_s}[\Pi_1(x) = g_1(x) \wedge \Pi_2(x) = g_2(x) \wedge \cdots \wedge \Pi_s(x) = g_s(x)] \geq 1 - \varepsilon/2.$$

Denote by $\mathcal{E}$ the event $\Pi_1(x) = g_1(x) \wedge \Pi_2(x) = g_2(x) \wedge \cdots \wedge \Pi_s(x) = g_s(x)$. We have for every $x \in \{-1,1\}^n$,

$$\begin{aligned}
\tilde{C}(x) &= \mathop{\mathbf{E}}_{\Pi_1,\Pi_2,\ldots,\Pi_s}[D(x)] \\
&= \mathbf{E}[D(x) \mid \mathcal{E}] \cdot \mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}] \cdot \mathbf{Pr}[\neg\mathcal{E}] \\
&= C(x) \cdot \mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}] \cdot \mathbf{Pr}[\neg\mathcal{E}].
\end{aligned}$$

On the one hand, we have

$$\tilde{C}(x) = C(x) \cdot \mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}] \cdot \mathbf{Pr}[\neg\mathcal{E}] \leq C(x) + \varepsilon/2.$$

On the other hand, we get

$$\tilde{C}(x) = C(x)\cdot\mathbf{Pr}[\mathcal{E}] + \mathbf{E}[D(x) \mid \neg\mathcal{E}]\cdot\mathbf{Pr}[\neg\mathcal{E}] \geq C(x)\cdot(1-\varepsilon/2) + (-1)\cdot(\varepsilon/2) \geq C(x) - \varepsilon.$$

This completes the proof of the claim. □

Now by Theorem 24 and Theorem 20, we have

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x)\cdot f(x)] \geq \varepsilon. \tag{1}$$

By the definition of $\tilde{C}$, Equation (1) implies that there exists some D, which is a formula whose leaves are functions with *deterministic* communication complexity at most R, such that

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[D(x)\cdot f(x)] \geq \varepsilon,$$

which implies

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[D(x) = f(x)] \geq 1/2 + \varepsilon/2.$$

Then by Theorem 21, there exists a function h, which can be expressed as the XOR of at most $O(\sqrt{s}\cdot\log(1/\varepsilon))$ leaf functions in D, such that

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[h(x)\cdot f(x)] \geq \frac{1}{s^{O(\sqrt{s}\cdot\log(1/\varepsilon))}},$$

which again implies

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[h(x) = f(x)] \geq \frac{1}{2} + \frac{1}{s^{O\left(\sqrt{s}\cdot\log(1/\varepsilon)\right)}}.$$

Finally, note that the k-party deterministic communication complexity of h is at most

$$O(R\cdot\sqrt{s}\cdot\log(1/\varepsilon)),$$

where $R = R^{(k)}_{\varepsilon/(2s)}(\mathcal{G})$. □

We are now ready to show Theorem 19.

Proof of Theorem 19. Consider Theorem 23 with f being GIP^k_n and $\mathcal{D}$ being the uniform distribution. Consider Theorem 22 with $\delta = 1/s^{O(\sqrt{s}\cdot\log(1/\varepsilon))}$. We have

$$O\left(R^{(k)}_{\varepsilon/(2s)}(\mathcal{G})\cdot\sqrt{s}\cdot\log(1/\varepsilon)\right) \geq n/(k4^k) - O\left(\sqrt{s}\cdot\log(s)\cdot\log(1/\varepsilon)\right),$$

which implies

$$s \geq \Omega\left(\frac{n^2}{k^2\cdot 16^k\cdot\left(R^{(k)}_{\varepsilon/(2n^2)}(\mathcal{G}) + \log n\right)^2\cdot\log^2(1/\varepsilon)}\right).$$ □

A Proofs of useful lemmas

A.1 Useful lemmas for formulas

The proofs in this section are essentially the same as that of [5].

Lemma 50 ([5], Theorem 20 restated)**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$, and let $f, C\colon \{-1,1\}^n \to \{-1,1\}$ be such that*

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x) = f(x)] \geq 1/2 + \varepsilon.$$

Let $\tilde{C}\colon \{-1,1\}^n \to \mathbb{R}$ be a ε-approximating function of C, i.e., for every $x \in \{-1,1\}^n$, $|C(x) - \tilde{C}(x)| \leq \varepsilon$. Then,

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x)\cdot f(x)] \geq \varepsilon.$$

Proof. Note that since $\tilde{C}$ ε-approximate C, we have for every $x \in \{-1,1\}^n$

$$\tilde{C}\cdot C(x) \geq 1-\varepsilon,$$

and

$$\tilde{C}\cdot(1-C(x)) \geq -1-\varepsilon.$$

Then,

$$\begin{aligned}
\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x)\cdot f(x)] &= \mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x)\cdot f(x) \mid C(x)=f(x)]\cdot \mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)=f(x)] \\
&\quad + \mathop{\mathbf{E}}_{x\sim\mathcal{D}}[\tilde{C}(x)\cdot f(x) \mid C(x)\neq f(x)]\cdot \mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)\neq f(x)] \\
&\geq (1-\varepsilon)\cdot \mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)=f(x)] + (-1-\varepsilon)\cdot\left(1-\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)=f(x)]\right) \\
&= 2\cdot \mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[C(x)=f(x)] - 1 - \varepsilon \\
&\geq 2\cdot(1/2+\varepsilon) - 1 - \varepsilon \geq \varepsilon,
\end{aligned}$$

as desired. □

Lemma 51 ([5], Theorem 21 restated)**.** *Let $\mathcal{D}$ be a distribution over $\{-1,1\}^n$ and let $\mathcal{G}$ be a class of functions. For $f\colon \{-1,1\}^n \to \{-1,1\}$, suppose that $D\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{FORMULA}[s]\circ\mathcal{G}$ is such that*

$$\mathop{\mathbf{Pr}}_{x\sim\mathcal{D}}[D(x) = f(x)] \geq 1/2 + \varepsilon_0.$$

Then there exists some $h\colon \{-1,1\}^n \to \{-1,1\} \in \mathsf{XOR}_{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)}\circ\mathcal{G}$ such that

$$\mathop{\mathbf{E}}_{x\sim\mathcal{D}}[h(x)\cdot f(x)] \geq \frac{1}{s^{O\left(\sqrt{s}\cdot\log(1/\varepsilon_0)\right)}}.$$

Proof. Let

$$D = F(g_1, g_2 \ldots, g_s)$$

be a device in $\mathsf{FORMULA}\circ\mathcal{G}$ where F is a formula and $g_1, g_2, \ldots, g_s$ are function from $\mathcal{G}$.

Let $p\colon \{-1,1\}^s \to \mathbb{R}$ be a ε_0-approximating polynomial for F of degree $d = O(\sqrt{s}\cdot\log(1/\varepsilon_0))$. Note that we can write

$$p(z) = \sum_{\substack{S\subseteq[s]:\\ |S|\leq d}} \hat{p}(S)\cdot\prod_{i\in S} z_i.$$

Also, for each $S \subseteq [s]$, we have

$$|\hat{p}(S)| = \left| \mathop{\mathbf{E}}_{z \in \{-1,1\}^s} [p(z) \cdot \prod_{i \in S} z_i] \right| \leq 1 + \varepsilon_0.$$

Now let

$$\tilde{D} := p(g_1, g_2 \ldots, g_s).$$

Note that $\tilde{D}$ is a ε_0-approximating function for D. Therefore, by Theorem 50, we have

$$\begin{aligned}
\varepsilon_0 &\leq \mathop{\mathbf{E}}_{x \sim \mathcal{D}} [D(x) \cdot f(x)] \\
&= \mathop{\mathbf{E}}_{x \sim \mathcal{D}} \left[\left(\sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} \hat{p}(S) \cdot \prod_{i \in S} g_i \right) \cdot f(x) \right] \\
&= \sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} \hat{p}(S) \cdot \mathop{\mathbf{E}}_{x \sim \mathcal{D}} \left[\prod_{i \in S} g_i \cdot f(x) \right] \\
&\leq \sum_{\substack{S \subseteq [s]: \\ |S| \leq d}} (1 + \varepsilon_0) \cdot \left| \mathop{\mathbf{E}}_{x \sim \mathcal{D}} \left[\prod_{i \in S} g_i \cdot f(x) \right] \right|.
\end{aligned}$$

The above equation is the sum of at most $s^{O(d)}$ summands. Therefore, there exists some $S \subseteq [s]$ such that

$$\left| \mathop{\mathbf{E}}_{x \sim \mathcal{D}} \left[\prod_{i \in S} g_i \cdot f(x) \right] \right| \geq \frac{\varepsilon_0}{(1 + \varepsilon_0) \cdot s^{O(d)}} \geq \frac{1}{s^{O\left(\sqrt{s} \cdot \log(1/\varepsilon_0)\right)}},$$

which implies that there exists some h, such that either $h = \prod_{i \in S} g_i$ or $h = -\prod_{i \in S} g_i$, and

$$\mathop{\mathbf{E}}_{x \sim \mathcal{D}} [h(x) \cdot f(x)] \geq \frac{1}{s^{O\left(\sqrt{s} \cdot \log(1/\varepsilon_0)\right)}}.$$

Finally, note that such h can be expressed as the XOR of at most d functions from $\mathcal{G}$. □

References

[1] Harry Buhrman, Ilan Newman, Hein Röhrig, and Ronald de Wolf. Robust polynomials and quantum algorithms. *Theory Comput. Syst.*, 40(4):379–395, 2007. doi.org/10.1007/s00224-006-1313-z.

[2] László Babai, Noam Nisan, and Mario Szegedy. Multiparty protocols, pseudorandom generators for logspace, and time-space trade-offs. *J. Comput. Syst. Sci.*, 45(2):204–232, 1992. doi.org/10.1016/0022-0000(92)90047-M.

[3] Eyal Kushilevitz and Noam Nisan. *Communication Complexity*. Cambridge University Press, 1997.

[4] Ben Reichardt. Reflections for quantum query algorithms. In *SODA*, pages 560–569, 2011. arXiv:1005.1601.

[5] Avishay Tal. The bipartite formula complexity of inner-product is quadratic. *Electronic Colloquium on Computational Complexity (ECCC)*, 23:181, 2016. ECCC:TR16-181.